Roman Kochnev

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Education

University of Würzburg, M.Sc. Computer Science (Informatik), Data Science specialization 2023 – 2026

- Focus: AutoML, Large Language Models, NLP
- Thesis: Enabling Symbolic Reasoning with Advanced RAG Systems

Lomonosov Moscow State University, M.Sc. Applied Mathematics and Computer Science 2020 – 2022

National University of Science and Technology MISIS, B.Sc. Applied Mathematics 2016 – 2020

Experience

Data Science Researcher (Master's Thesis), Bosch Center for AI (BCAI) – Renningen, Germany Jan 2026 – Jul 2026

- Developed **APOLO**, casting ontology learning as prompt optimization: LLM-based ontology learners tuned with DSPy (GEPA) without weight updates outperform SOTA baselines; accepted at ISWC 2026
- Built **GRAIL**, a reasoning-focused QA benchmark for evaluating RAG and GraphRAG systems on OWL ontologies, with reasoner-verified ground truth across 1500+ questions
- Designed **OWL2Doc**, a multi-agent pipeline (Planner, Writer, Critic) generating annotated text-ontology pairs from expert-curated OWL ontologies

Research Assistant (HiWi), University of Würzburg – Würzburg, Germany Jun 2025 – Present

- First author of **NNGPT** (CVPRW 2026), an AutoML framework where an LLM generates PyTorch architectures, tunes hyperparameters and self-improves via RL feedback
- Co-developed the original **LEMUR** neural network dataset; co-author of **LEMUR 2** (CVPRW 2026)
- First author of **Optuna vs Code Llama** (ICCVW 2025), benchmarking LLM-based hyperparameter tuning against Optuna

Software Developer and Analyst, ROSBANK, formerly Société Générale Group – Moscow May 2022 – Sep 2023

- Built and maintained credit scoring models
- Automated lending processes and delivered analyses, reports and visualizations that informed business decisions
- Sourced, cleaned and validated data from closed internal banking systems
- Partnered with cross-functional teams to turn company data into business solutions; mentored interns and junior colleagues

Data Analyst, GlowByte Consulting – Moscow Sep 2021 – Apr 2022

- Developed and optimized PL/SQL scripts, built Power BI dashboards and ran A/B tests for analytics projects across top-tier banks and major food retail chains

Publications

1. **R. Kochnev**[†], H. T. Mai[†], C. X. Chu, L. Lange, H. Paulheim, D. Stepanova, "APOLO: Automatic Prompt Optimization for Ontology Learning", *ISWC*, 2026. ([†]equal contribution)
2. **R. Kochnev**, W. Khalid, T. Uzun, X. Zhang, Y. Dhameliya, F. Qin, C. Vysyaraju, R. Duvvuri, A. Goyal, D. Ignatov, R. Timofte, "NNGPT: Rethinking AutoML with Large Language Models", *CVPRW*, 2026.
3. T. Uzun, W. Khalid, S. Din, S. Mulukuledu, A. Singh, C. Vysyaraju, R. Duvvuri, A. Goyal, Y. R. Lukhi, M. A. Hussain, K. Jesani, U. Shrestha, Y. Mittal, **R. Kochnev**, P. Kadam, M. Ikram, H. Moradiya, A. Arslanian, D. Ignatov, R. Timofte, "LEMUR 2: Unlocking Neural Network Diversity for AI", *CVPRW*, 2026.
4. A. Goodarzi, **R. Kochnev**, W. Khalid, H. Goudarzi, F. Qin, T. Uzun, Y. Dhameliya, Y. Kathiriya, Z. Bentyn, D. Ignatov, R. Timofte, "LEMUR Neural Network Dataset: Towards Seamless AutoML", *arXiv*, 2025.
5. **R. Kochnev**, A. Goodarzi, Z. Bentyn, D. Ignatov, R. Timofte, "Optuna vs Code Llama: Are LLMs a New Paradigm for HPO?", *ICCVW*, 2025.
6. **R. Kochnev**, R. Senchenko, "Intelligent Method for Real Estate Price Prediction in Moscow Using a Machine Learning Model" (in Russian), *Proceedings of the 75th Days of Science of NUST MISIS Students*, 2020. ISBN 978-5-907227-11-8.

Technical Skills

Programming Languages: Python, SQL, Java, C++

ML / Deep Learning: PyTorch, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, OpenCV

LLM & AutoML: Fine-tuning (PEFT/LoRA), Prompt Optimization (DSPy, GEPA), RAG, Ontology Reasoning, Neural Architecture Search

MLOps & Infra: Docker, Azure, MLflow, Kubernetes, Git

Tools: Jira, Confluence, Power BI, Excel, LaTeX

Languages: English (C1), German (B1), Russian (Native)